

# ADAT v2.0 Converter — Functional Requirements

**Version:** 1.0-draft **Updated:** 2026-06-16 DRAFT

**Applies to:** SomaDataIO (R) and Canopy/somadata (Python) **Format ref:**  
`adat_v2.0_spec.md`

## 1. Overview and Purpose

Specifies requirements for `to_v2_adat()`, to be added to [SomaDataIO](#) and [Canopy](#). Converts one or more pre-v2.0 ADAT files or in-memory objects into a single v2.0 combined-format ADAT. The v2.0 format is defined in `adat_v2.0_spec.md` (authoritative reference for all field definitions).

### 1.1 Motivation and Release Priority

The primary near-term use case is an Illumina Connected Multiomics (ICM) feature to bridge array-based SomaScan results into NGS-based SomaSeq RFU space. The functionality is expected to be present in Canopy/SomaData and available in Illumina's Artifactory registry. The v2.0 spec is expected to be released publicly through both SomaDataIO and Canopy/Somadata via GitHub, CRAN, and PyPI, however public release will follow a the internal ICM release to Artifactory, timeline TBD.

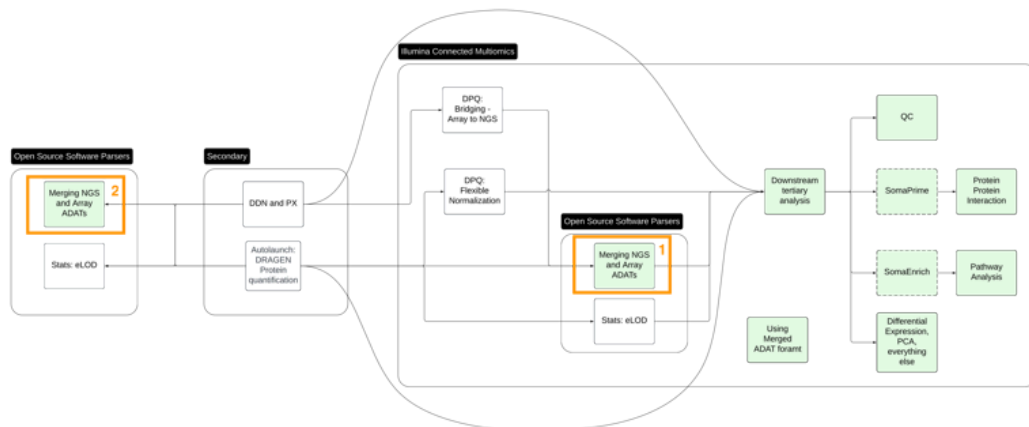
### 1.2 Scope

**In scope:** single-input conversion; two-input merge; all approved paths in Section 4; writer updates for v2.0 output.

**Out of scope:** back-conversion from v2.0; more than two simultaneous inputs; array AssayVersion below v4.

### 1.3 ICM user workflow

See figure below showing the overall user workflow. Orange boxes represent the parser role in the workflow with numbers representing implementation priority.



## 2. Terminology and Input Types

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| Type Label                    | Short Name                 | Detection Criteria                                                                                                                                       |
|-------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-v2.0 native array         | <code>native_array</code>  | <code>SlideId</code> + <code>Subarray</code> non-empty; <code>ProcessSteps</code> do NOT end with bridged terminal; <code>AssayVersion</code> v4+        |
| Pre-v2.0 array bridged to NGS | <code>bridged_array</code> | <code>SlideId</code> + <code>Subarray</code> non-empty; <code>ProcessSteps</code> end with bridged terminal (Section 2.2); <code>AssayVersion</code> v4+ |
| Pre-v2.0 native NGS           | <code>native_ngs</code>    | <code>SOMAmxReads</code> column present and non-empty; <code>SlideId</code> / <code>Subarray</code> absent or empty                                      |
| v2.0 combined                 | <code>v2_combined</code>   | Header <code>FileVersion</code> == "2.0"                                                                                                                 |

### 2.2 Bridged Array Terminal ProcessSteps

Classified as `bridged_array` when `ProcessSteps` ends with (case-sensitive, last 3 in order):

1 | `CrossPlatformPlateScaling`, `CrossPlatformCalibrate`, `MedNormExt`

Canonical example ( `bridged_array_2_NGS.adat` ):

```
1 Raw RFU, Hyb Normalization, medNormInt (SampleId), plateScale, Calibration, anmlQC, qcCheck,
  anmlSMP CrossPlatformPlateScaling, CrossPlatformCalibrate, MedNormExt
```

Only the terminal 3 steps are checked. Earlier steps may vary. `FileVersion == "2.0"` is checked first; if present all other detection is skipped. Array inputs below AssayVersion v4 are rejected (Section 5.3).

### 3. Function Signature

```
1 # Python
2 def to_v2_adat(adats: list[str | Adat], med_norm_ref: str | None = None) -> Adat
```

```
1 # R
2 to_v2_adat <- function(adats, med_norm_ref = NULL)
```

**adats:** list of 1-2 items; each is a file path or in-memory adat object (mixed allowed). Paths read via existing `read_adat()`.

**med\_norm\_ref:** optional MedNorm reference ID (Section 8).

**Returns:** single v2.0 adat object. Caller writes via existing `write_adat()`.

**Raises:** `ValueError` (Python) / `rlang::abort()` (R) for Section 5 conditions.

```
1 # Python
2 result = somadata.to_v2_adat(["bridged_array_2_NGS.adat", "ngs_sample.adat"])
3 somadata.write_adat(result, "output_v2.adat")
```

```
1 # R
2 result <- to_v2_adat(list("bridged_array_2_NGS.adat", "ngs_sample.adat"))
3 write_adat(result, "output_v2.adat")
```

### 4. Approved Conversion and Merge Paths

All paths must operate within the same sample matrix type (Section 5.2, Appendix B). Any unlisted combination must be rejected.

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| #   | Input A       | Input B      | Output AssayType | Notes                                                 |
|-----|---------------|--------------|------------------|-------------------------------------------------------|
| 1** | bridged_array | native_nginx | Mixed            | <b>Primary use case.</b> Common MedNorm ref required. |
| 2   | bridged_array | v2_combined  | Mixed            | v2_combined may be Array/NGS/Mixe                     |

|   |               |              |              |                                                  |
|---|---------------|--------------|--------------|--------------------------------------------------|
|   |               |              |              | d. MedNorm validated.                            |
| 3 | native_ngs    | v2_combined  | Mixed or NGS | NGS if v2_combined is NGS-only; Mixed otherwise. |
| 4 | native_array  | native_array | Array        | Both must be same SomaScan AssayVersion.         |
| 5 | v2_combined   | v2_combined  | Derived      | AssayType from union of SampleReadout values.    |
| 6 | bridged_array | none         | Array        | Single-input conversion only.                    |
| 7 | native_array  | none         | Array        | Single-input conversion only.                    |
| 8 | native_ngs    | none         | NGS          | Single-input conversion only.                    |

\*\* Primary use case for ICM. This path should be prioritized in case scope need to be cut due to timeline.

## 5. Error Conditions

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| Sec | Condition                          | Error Message                                                                                            |
|-----|------------------------------------|----------------------------------------------------------------------------------------------------------|
| 5.1 | native_array + any NGS-space input | Native array data cannot be combined with NGS-space data. Array must be bridged first. Input: {filename} |
| 5.1 | Unlisted combination               | Unsupported input combination: {typeA} +                                                                 |

|     |                                                                           |                                                                                                                          |
|-----|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|     |                                                                           | <code>{typeB}</code> .                                                                                                   |
| 5.1 | Single <code>v2_combined</code> input                                     | Warning (not error): <i>Input already v2.0. Returning as-is.</i>                                                         |
| 5.1 | Empty <code>adats</code> list                                             | <i>adats must contain at least one ADAT.</i>                                                                             |
| 5.1 | More than 2 inputs                                                        | <i>to_v2_adat() accepts max 2 inputs. Received: {n}.</i>                                                                 |
| 5.2 | Incompatible <code>SampleMatrix</code> (Sample rows only; see Appendix B) | <i>SampleMatrix mismatch: '{matrixA}' vs '{matrixB}'. 'Plasma' and 'EDTA Plasma' are equivalent.</i>                     |
| 5.3 | Array <code>AssayVersion</code> < v4                                      | <i>AssayVersion '{version}' not supported. Minimum is v4.</i>                                                            |
| 5.4 | <code>Ref.MedNormExt</code> vectors not identical across sources          | <i>MedNorm external reference mismatch. Provide med_norm_ref to specify a reference.</i>                                 |
| 5.4 | <code>med_norm_ref</code> value not found                                 | <i>med_norm_ref '{value}' not found in Ref.MedNorm.Id values.</i>                                                        |
| 5.5 | NGS+NGS: <code>ProcessSteps</code> not identical                          | <i>NGS ProcessSteps mismatch: '{stepsA}' vs '{stepsB}'.</i>                                                              |
| 5.6 | <code>bridged_array</code> terminal steps mismatch                        | <i>Array input not fully bridged. Steps must end with CrossPlatformPlateScaling, CrossPlatformCalibrate, MedNormExt.</i> |
| 5.6 | <code>native_ngs</code> steps don't match required sequence (Appendix A)  | <i>NGS ProcessSteps don't match required sequence for bridged array merge.</i>                                           |

## 6. Platform Detection Logic

```
1 for each ADAT in adats:
2     if FileVersion == "2.0": classify as v2_combined
3     elif SlideId and Subarray non-empty:
4         if AssayVersion < v4: raise error 5.3
5         if ProcessSteps ends with [CrossPlatformPlateScaling, CrossPlatformCalibrate,
MedNormExt]:
6             classify as bridged_array
7             else: classify as native_array
8     elif SOMAmerReads column present and non-empty: classify as native_ngs
9     else: raise error: unrecognized format
10 then look up combination in Section 4; raise error if unlisted
```

## 7. Field Conversion Requirements

Complete field-by-field mappings are in `adat_v2.0_spec.md` Sections 3.2 (Array), 3.3 (NGS), 3.4 (Mixed). Key categories:

- **Header JSON consolidation:** legacy per-plate flat fields (`PlateScale_Scalar_<PlateId>` etc.) consolidated into JSON keyed by `PlateId` with `PlatformSpecific` / `CrossPlatform` sub-keys for NGS.
- **ProcessSteps/ReportConfig:** reformatted to JSON `{"<id>": ["step1", ...]}`; distinct sets get unique IDs on merge; samples mapped via `ProcessStepsId` / `ReportConfigId`.
- **New generated fields:** `SampleReadout`, `UniqueSampleKey` (GUID per sample, immutable), `SourceFileId`, `ProcessStepsId`, `ReportConfigId`, `HybNormStatus` (PASS if scale 0.4-2.5), `MedNormIntStatus`.
- **COL\_DATA renames:** `EntrezGeneID` → `EntrezGeneId`;  
`Cal_<PlateId>` → `PlatformSpecificCalibrate_<PlateId>_ScaleFactor`;  
`CalQcRatio_<PlateId>_<QCSampleId>` → `QCRatio_<PlateId>`; reference fields get `Ref.Array.` or `Ref.NGS.` prefix; NGS-specific fields get `_NGS` suffix; `HybControl` generated from `Type` field. Removed: `SeqIdVersion`, `SomaId`, `ColCheck`, `Units`.
- **ROW\_DATA renames (array):** `PlatePosition` → `WellPosition`;  
`HybControlNormScale` → `HybNormScaleFactor`; `RowCheck` → `RowCheckStatus`;  
`StudyId` → `Project`; `SubjectID` → `SubjectId`; `Barcode2d` → `MatrixTubeBarcode`;  
`GeneratedBy` header → `SoftwareVersion` sample row; `CreatedDate` header → `PlateRunDate` if blank. Removed: `ExtIdentifier`, `SsfExtId`, `ScannerID`, `Barcode`, `SampleName`, `SampleNotes`, `AliquotingNotes`, `AssayNotes`, `SampleDescription`, `TimePoint`, `SampleGroup`, `SiteId`, `CLI`, `PercentDilution`.
- **ROW\_DATA renames (NGS):** PascalCase normalization for all IDs;  
`HybNorm_1_ScaleFactor` → `HybNormScaleFactor`; all `*_PassFlag` → `*Status`;  
header fields `InstrumentType`, `Flowcell`, `YieldDemux`, `YieldQ30Demux`,

Q30WeightedMean moved to sample rows (replicated to all NGS rows);  
ProbePlate → NGSPlateMasterMixLot ; MatrixType → KitType .

## 8. MedNorm Reference Handling

When merging ADATs where either source has MedNormExt in ProcessSteps, the Ref.MedNormExt.<Matrix> reference RFU vectors (COL\_DATA) must be element-wise identical across all sources for shared SeqIds. If mismatched: raise error 5.4 unless med\_norm\_ref is specified, then use values from the source whose Ref.MedNorm.Id matches. For NGS+NGS merges, ProcessSteps must also be identical. Single-input conversions skip all MedNorm checks.

## 9. SeqId Union and Missing Value Fill

Output contains the union of all SeqIds from both sources. SeqIds absent in a source get NA in that source's sample rows. COL\_DATA is populated for all SeqIds in the union; platform-specific fields for the other platform are blank/NA. Column order may be by SeqId ascending; ordering must be consistent between COL\_DATA and data matrix.

## 10. Header Metadata Merge Rules

The following header fields are merged as pipe-delimited lists of unique values: Title , StudyOrganism , StudyMatrix , SOMAmerReferenceSource .

All plate-keyed JSON fields are merged by combining all PlateId keys from both sources. If both sources contribute the same PlateId, raise an error.

## 11. Writer Requirements

The existing write\_adat() in both packages must be updated to support v2.0 output. Version dispatch: if FileVersion == "2.0" use v2.0 writer path; otherwise existing pre-v2.0 path (unchanged).

**v2.0 writer rules:** closed header field set (all spec Section 2.3 fields present; no additional); no ! prefix on field names; no !Checksum line; JSON fields serialized as single-line minified JSON; field values comply with spec Section 2.2 types; ASCII TAB-delimited encoding; sections in order ^HEADER , ^COL\_DATA , ^ROW\_DATA , ^TABLE\_BEGIN ; COL\_DATA and ROW\_DATA each have Name and Type rows.

## 12. Example Files

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| File | Type | Role |
|------|------|------|
|------|------|------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |               |                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <a href="https://ilmn.sharepoint.com/:u:r/teams/Cassini/Shared%20Documents/06_Bioinformatics_SW/Tiger%20Teams/ADAT%20Format%20and%20Parsers/Array%20ADAT%20Examples/example_array_ADATs/dragen_protein_project_CSN106_Step1_Bridged.adat?csf=1&amp;web=1&amp;e=ePMtbU">https://ilmn.sharepoint.com/:u:r/teams/Cassini/Shared%20Documents/06_Bioinformatics_SW/Tiger%20Teams/ADAT%20Format%20and%20Parsers/Array%20ADAT%20Examples/example_array_ADATs/dragen_protein_project_CSN106_Step1_Bridged.adat?csf=1&amp;web=1&amp;e=ePMtbU</a>                                                                                                                                                         | bridged_array | Primary input. ProcessSteps ends with CrossPlatformPlateScaling, CrossPlatformCalibrate, MedNormExt. AssayVersion v5.0, EDTA Plasma. |
| <a href="https://ilmn.sharepoint.com/:u:r/teams/Cassini/Shared%20Documents/06_Bioinformatics_SW/Tiger%20Teams/ADAT%20Format%20and%20Parsers/Array%20ADAT%20Examples/example_array_ADATs/SS-26158660_v5.0_EDTAPlasma.bridge.hybNorm.medNormInt.plateScale.leakDetection.calibrate.anmlQC.qcheck.anmlSMP.20260220.adat?csf=1&amp;web=1&amp;e=vErBAO">https://ilmn.sharepoint.com/:u:r/teams/Cassini/Shared%20Documents/06_Bioinformatics_SW/Tiger%20Teams/ADAT%20Format%20and%20Parsers/Array%20ADAT%20Examples/example_array_ADATs/SS-26158660_v5.0_EDTAPlasma.bridge.hybNorm.medNormInt.plateScale.leakDetection.calibrate.anmlQC.qcheck.anmlSMP.20260220.adat?csf=1&amp;web=1&amp;e=vErBAO</a> | native_array  | Native array.                                                                                                                        |
| <a href="https://ilmn.sharepoint.com/:u:r/teams/Cassini/Shared%20Documents/06_Bioinformatics_SW/Tiger%20Teams/ADAT%20Format%20and%20Parsers/NGS%20ADAT%20Examples/HRC_Plasma_Sample_Handling_v2_dragen-protein-quant_Step3_SampleNor">https://ilmn.sharepoint.com/:u:r/teams/Cassini/Shared%20Documents/06_Bioinformatics_SW/Tiger%20Teams/ADAT%20Format%20and%20Parsers/NGS%20ADAT%20Examples/HRC_Plasma_Sample_Handling_v2_dragen-protein-quant_Step3_SampleNor</a>                                                                                                                                                                                                                           | native_ngs    | NGS input. AssayVersion Illumina Protein Prep 9k, Plasma.                                                                            |



|                                                                                                           |             |                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">m.adat?csf=1&amp;web=1&amp;e=0HA</a><br>OIV <a href="#">Connect your OneDrive a</a><br>ccount |             |                                                                                                                                |
| TBD when available:<br>example_v2.0_Mixed.adat                                                            | v2_combined | Target output for path 1<br>(bridged_array + native_ngs).<br>Shows v2.0 header,<br>COL_DATA, ROW_DATA, and<br>JSON formatting. |

## 12.1 Primary Use Case Walkthrough

**Input A:** bridged\_array\_2\_NGS.adat | Type: bridged\_array | AssayVersion: v5.0 | StudyMatrix: EDTA Plasma

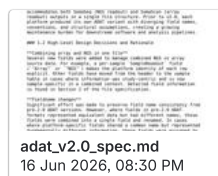
ProcessSteps: Raw RFU, Hyb Normalization, medNormInt (SampleId), plateScale, Calibration, anmlQC, qcCheck, anmlSMP  
CrossPlatformPlateScaling, CrossPlatformCalibrate, MedNormExt

**Input B:** HRC\_Plasma\_Sample\_Handling\_v2\_...SampleNorm.adat | Type: native\_ngs | AssayVersion: Illumina Protein Prep 9k | StudyMatrix: Plasma (equivalent to EDTA Plasma)  
ProcessSteps: Raw, HybNorm, MedNormInt, PlatformSpecificPlateScale, PlatformSpecificCalibrate, CrossPlatformPlateScale, CrossPlatformCalibrate, MedNormExt

**Expected output:** example\_v2.0\_Mixed.adat | AssayType: Mixed | FileVersion: 2.0 | Array and NGS sample rows with SampleReadout set accordingly.

## Appendix A: Draft ADAT v2.0 Spec

Markdown draft of ADAT v2.0 Spec (under review)



## Appendix B: Required ProcessSteps Sequences

**Path 1 — bridged array terminal (last 3 steps required):**

1 | CrossPlatformPlateScaling, CrossPlatformCalibrate, MedNormExt

**Path 1 — native NGS full sequence required:**

1 Raw, HybNorm, MedNormInt, PlatformSpecificPlateScale, PlatformSpecificCalibrate, CrossPlatformPlateScale, CrossPlatformCalibrate, MedNormExt

Both defined in `adat_v2.0_spec.md` Section 3.4. For NGS+NGS merges (Path 5): ProcessSteps must be identical across both sources (no predefined sequence required). For Array+Array (Path 4): no ProcessSteps restriction beyond classification rules.

## Appendix C: SampleMatrix Equivalence Rules

Matrix check (Section 5.2) uses equivalence groups below. Comparison is case-insensitive. Only `SampleType="Sample"` rows used; Calibrators/QCs/Buffers excluded.

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| Canonical Name | Equivalent Values         |
|----------------|---------------------------|
| Plasma         | Plasma , EDTA Plasma      |
| Serum          | Serum                     |
| CSF            | CSF , Cerebrospinal Fluid |

Values outside these groups are treated as distinct.